RESEARCH PROJECT REPORT

PROJECT #78:

AI-USE DISCLOSURE GOVERNANCE AND CONTRIBUTOR COMPLIANCE IN OPEN-SOURCE SOFTWARE

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ABSTRACT

Generative artificial intelligence is changing how developers contribute to open-source software (OSS), creating questions about disclosure, responsibility, and review. This study examines how OSS projects govern AI-assisted contributions, the extent of observable disclosure in pull requests (PRs), and how projects communicate and enforce their expectations. We combined rule-based analysis of 41,338 eligible PRs from 38 repositories with manual coding of contribution policies. The detector achieved 99.3% accuracy on 708 manually reviewed PRs that also informed its refinement. Observable disclosure appeared in 11,598 PRs (28.1%), with repository-level rates ranging from 0% to 88.3%. Policy visibility was positively associated with disclosure rates (Spearman's $\rho = .618$, $p < .001$). Governance requirements extended beyond disclosure to human verification, acceptable use, and stated consequences; reviewer workload was the most frequently stated rationale. These findings suggest that clear prompts within contribution workflows can support communication of AI-use expectations. However, recorded disclosure does not measure all AI use or establish compliance with every policy, and documented consequences do not demonstrate actual enforcement. The study provides a baseline for evaluating how governance supports transparent contributions and sustainable review practices.

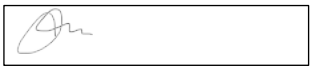
SIGNED STATEMENT OF CONTRIBUTION

Student

I hereby declare that:

1. This report is the result of the final year project work carried out by my project partner (see cover page) and I under the guidance of our supervisor (see cover page) in the 2026 academic year at the Department of Electrical, Computer, and Software Engineering, Faculty of Engineering and Design, University of Auckland.
2. This report is not the outcome of work done previously.
3. This report is not the outcome of work done in collaboration, except that with a potential project sponsor (if any) as stated in the text.
4. This report is not the same as any report, thesis, conference article or journal paper, or any other publication or unpublished work in any format.

In the case of a continuing project, please state clearly what has been developed during the project and what was available from previous year(s):

Signature: 

Date: 18 October 2026

Supervisor

I confirm that the project work undertaken by this student in the 2026 academic year is / is not (strikethrough as appropriate) part of a continuing project, components of which have been completed previously. Comments, if any:

Signature: 

Date: 19 Oct 2026

Table of Contents

Acknowledgements **6**

List of Figures **7**

List of Tables **8**

Glossary of Terms **9**

Abbreviations **10**

1. INTRODUCTION **1**

1.1 Background 1

1.2 Problem statement 1

1.3 Research questions 2

1.4 Objectives 2

1.5 Scope of the study 2

1.6 Significance of the research 3

2. LITERATURE REVIEW **3**

2.1 OSS contribution models and governance 3

2.2 Contributor workflows and maintainer responsibilities 4

2.3 AI-assisted software development 4

2.4 Quality, accountability, and review challenges 4

2.5 AI-use disclosure and its measurement in OSS 5

2.6 AI governance policies, visibility, and enforcement 6

2.7 Research gap and study contribution 7

3. METHODOLOGY **8**

3.1 Research design 8

3.2 Repository identification 8

3.3 Repository eligibility criteria 8

3.4 Automated analysis tool 8

3.5 Automated disclosure classification 9

3.6 Inter-rater reliability sample selection 10
3.7 Inter-rater reliability 11
3.8 Expanded detector evaluation 11
3.9 Full repository analysis 12
3.10 Visibility score 12
3.11 Manual governance coding 13
3.12 Statistical analysis 13
3.13 Data analysis 14
4. QUANTITATIVE RESULTS 15
4.1 Detector performance on the development sample 15
4.2 Repository disclosure results 16
4.3 Types of AI-use disclosures 17
4.4 Visibility and disclosure rate 18
5. GOVERNANCE ANALYSIS RESULTS 19
5.1 Governance requirements and practices 19
5.2 Enforcement mechanisms and stated consequences 20
5.3 Motivations behind AI governance policies 20
6. DISCUSSION 22
6.1 Interpreting disclosure and compliance 22
6.2 Policy visibility and contribution workflows 23
6.3 Human responsibility and stated consequences 23
6.4 Motivations underlying AI governance 24
6.5 Limitations of the study 24
6.6 Synthesis of answers to the research questions 25
6.7 Practical implications and future research 26
7. CONCLUSION 26
References 28
Appendix A 30

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List of Figures

Figure 1: AI-Use Disclosure Rates Across OSS Repositories 17
Figure 2: Distribution of AI-Use Disclosure Types Across OSS Pull Requests 18
Figure 3: Relationship Between AI-Disclosure Policy Visibility and Disclosure Rates 18
Figure 4: Distribution of Stated Consequences Across OSS Repositories 20
Figure 5: Stated Motivations for AI-Use Governance 22

List of Tables

Table 1: Repository Selection Requirements 8
Table 2:Representative AI-use disclosure detection rules 10
Table 3: Visibility Scoring 12
Table 4: Detector-Validation confusion matrix 15
Table 5: Detector-performance metrics..... 15

Glossary of Terms

AI-assisted contribution	A submitted change developed with support from an AI tool; assistance may involve code, tests, documentation, or related work.
Ambiguous disclosure	An AI-related response whose meaning does not clearly establish use or non-use, such as a general policy acknowledgement.
Contribution governance	The rules, responsibilities, decision processes, and consequences used to coordinate and assess project contributions.
Disclosure compliance	Satisfaction of the disclosure rule applicable to a contribution; establishing it requires knowing the rule and whether its conditions apply.
Disclosure evidence	Text or a completed field that the study's coding rules recognise as a r response concerning AI use.
Eligible pull request	A human-authored, closed PR that meets the study's time-window and other inclusion criteria.
Enforcement provision	A documented response to a policy violation, such as rejection or restrictions on future participation; distinct from an observed action.
Maintainer	A project participant responsible for reviewing, integrating, or supporting contributions and sustaining the project.
Negative disclosure	An explicit statement that AI was not used; the term describes reported use rather than detector performance.
Observed disclosure rate	The proportion of analysed eligible PRs containing disclosure evidence recognised by the detector.
Policy visibility	The accessibility of an AI-use policy within the contribution workflow, scored from its documented location.
Positive disclosure	A statement reporting AI assistance; distinct from a positive classification for the presence of any disclosure.
Provenance	The origin and traceability of contributed material, including information relevant to ownership and licensing.
Pull request	A proposal to merge changes into a repository, normally accompanied by a description and review process.
Purposive sample	A sample selected because its members meet criteria relevant to the research, rather than to represent the whole population randomly.
Rule-based detector	Software that classifies text using predefined matching patterns and structural rules.

Abbreviations

AI	Artificial Intelligence
AIL	Numeric AI-use disclosure field with repository-specific conventions
API	Application programming interface
DCO	Developer Certificate of Origin
F1	Harmonic mean of precision and recall
FN	False Negative
FP	False Positive
GenAI	Generative artificial Intelligence
GPL	GNU General Public License
HTML	HyperText Markup Language
IRR	Inter-rater reliability
MIT	Massachusetts Institute of Technology, as used in the MIT License
OSS	Open-source Software
PR	Pull request
REST	Representational State Transfer
RQ	Research Question
TN	True negative
TP	True Positive
TRACE	Transparency, Responsibility, Attribution, Constraints and Enforcement
UTC	Coordinated Universal Time

1. INTRODUCTION

1.1 Background

Open-source software (OSS) plays a fundamental role in modern software development, underpinning many widely used technologies, applications, and digital services. OSS projects rely on contributions from diverse communities, including independent volunteers and corporate-supported developers, whose motivations range from personal interest and skill development to commercial and organisational objectives (Riehle, 2007; Zhang et al., 2021). The collaborative nature of OSS development depends on established contribution workflows, in which maintainers review submitted changes to ensure they meet project standards for quality, security, and maintainability.

Recent advances in artificial intelligence (AI), particularly generative AI tools such as GitHub Copilot and ChatGPT, are changing how software is developed. These tools can assist with tasks such as code generation, debugging, testing, and documentation, although their effectiveness varies depending on the task and development context (Becker et al., 2025; Pereira et al., 2025). While AI-assisted development offers opportunities to improve developer productivity, it also introduces concerns regarding code quality, intellectual property, transparency, and contributor accountability.

These concerns are particularly relevant to OSS projects, where maintainers must evaluate contributions from individuals with varying levels of experience and familiarity with the codebase. AI-generated or AI-assisted contributions may introduce additional review demands, particularly when contributors cannot adequately explain, verify, or take responsibility for the submitted changes. In response, OSS communities are increasingly adapting their contribution guidelines to address the use of AI, including requirements for contributors to disclose AI assistance in pull requests (PRs).

1.2 Problem statement

Existing research has examined the productivity and code-quality implications of AI-assisted development, while emerging studies investigate how OSS communities govern its use (Chen et al., 2026; Yang et al., 2026). In response to concerns surrounding transparency, accountability, and code quality, OSS projects have introduced policies requiring AI-use disclosure, verification, or restrictions on certain AI-assisted contributions.

However, the presence of a policy does not necessarily indicate how effectively its requirements are communicated or reflected in contributor behaviour. Policies vary in visibility, enforcement mechanisms, and expectations placed on contributors. Furthermore, identifying actual AI use is challenging, as contributors may not disclose their use of AI, while the absence of a disclosure does not establish that AI was used.

This research addresses these challenges by examining documented AI-use governance practices alongside observable disclosure in PR descriptions. By analysing policy visibility, disclosure frequency, and documented enforcement mechanisms, the study investigates how OSS projects respond to AI-assisted development without assuming that recorded disclosures represent all instances of AI use.

1.3 Research questions

This study investigates how OSS projects are adapting their contribution governance practices to address AI-assisted development, with particular emphasis on disclosure requirements and their implementation.

The research is guided by three questions:

RQ1: How are OSS projects adapting contribution governance practices in response to AI-assisted development?

RQ2: To what extent do contributors comply with mandatory AI-use disclosure policies in OSS pull requests?

RQ3: How do OSS projects operationalise and enforce AI-use disclosure policies within contribution workflows?

In RQ2, compliance is operationalised as observable disclosure: the proportion of eligible PRs containing a detector-recognised AI-use statement. This measures what contributors record in PR descriptions, not whether each contribution met every applicable policy requirement.

Together, these questions examine the development of AI-related governance practices, the extent of observable disclosure within PRs, and the mechanisms through which projects communicate and implement their requirements.

1.4 Objectives

The primary objective of this study is to investigate how OSS projects govern AI-assisted contributions and how disclosure requirements are reflected in observable contribution records.

Specifically, the research aims to:

1. Characterise AI-use governance policies by examining disclosure requirements, contributor responsibilities, restrictions, and policy rationales across selected repositories.
2. Develop and validate an automated disclosure analyser to identify explicit AI-use and non-use statements in GitHub PR descriptions, using manually labelled samples to assess detection performance.
3. Measure observable disclosure rates across eligible PRs and examine their relationship with policy visibility, assessed using a defined four-level visibility rubric.
4. Analyse governance implementation through documented PR templates, contribution guidelines, enforcement mechanisms, and stated consequences.

1.5 Scope of the study

This study examines 38 selected GitHub repositories and 41,338 eligible human-authored PRs closed within a four-month analysis window. Repositories were purposively selected based on identifiable AI-use governance requirements.

A mixed-methods approach combines automated classification of AI-use disclosure statements in PR descriptions with manual coding of publicly available governance policies. Disclosure rates are calculated as the proportion of eligible PRs containing identifiable disclosures, while policy visibility is evaluated using a four-level rubric (0–3). Governance mechanisms, contributor responsibilities, and policy rationales are examined through predefined coding categories.

The study focuses on observable disclosure and documented governance, rather than determining actual AI use or directly evaluating code quality. Consequently, disclosure rates should not be interpreted as definitive measures of policy compliance, and documented enforcement mechanisms do not necessarily indicate that enforcement actions occurred.

1.6 Significance of the research

As AI-assisted development becomes increasingly common, OSS communities face challenges in maintaining transparency, accountability, and contribution quality. Understanding how projects establish and implement governance requirements can help inform their responses to these challenges.

This research contributes empirical evidence by connecting documented AI-use policies with observable disclosure practices across OSS repositories. Its findings may help maintainers improve policy visibility, clarify contributor expectations, and integrate disclosure requirements into contribution workflows.

Additionally, the study provides a reproducible approach to analysing AI-use disclosures and governance practices, supporting future research into the relationship between OSS policy design and contributor behaviour.

2. LITERATURE REVIEW

This review brings together research on OSS participation, AI-assisted development, code quality, and contribution governance. These strands explain why the effects of AI depend on collaborative workflows and why disclosure has become a subject of project policy.

2.1 OSS contribution models and governance

Open-source software encompasses voluntary participation and commercially supported development. Riehle (2007) distinguishes community and commercial models, while Zhang et al. (2021) demonstrate substantial, uneven corporate participation in OpenStack. Li et al. (2024) synthesise varied commercial motivations and contribution models. Together, these studies indicate that OSS participation cannot be understood solely as unpaid, decentralised work.

Across these arrangements, collaboration requires mechanisms for coordinating contributions and deciding what enters a shared codebase. Pull-request review allows maintainers to assess proposed changes before integration (Gousios et al., 2014), while contribution guidelines communicate expected procedures and standards (Elazhary et al., 2019). Such mechanisms are particularly relevant when contributors have different affiliations, priorities and levels of project familiarity, and maintenance responsibilities are distributed across participants.

Governance also concerns how decisions are made and justified. Sharma et al. (2024) identify multiple decision rationales in Python, including core-developer consensus and decisions by designated leaders. This highlights that open participation does not imply uniform decision-making authority. For AI-assisted contributions, existing guidelines and review processes therefore provide institutional settings for specifying disclosure, verification and responsibility requirements (Chen et al., 2026). Examining these policies situates AI governance within established practices for coordinating participation and evaluating contributions.

2.2 Contributor workflows and maintainer responsibilities

Pull-based development separates the preparation and submission of changes from decisions about their integration. Gousios et al. (2014) show how pull requests support iterative review, feedback and revision before acceptance. Contributors propose changes, while maintainers oversee their assessment and incorporation into the shared codebase. Setia et al. (2012) associate participation beyond a project's formal development team with higher software quality, reinforcing the importance of processes that accommodate wider participation.

These workflows create complementary responsibilities. Contributors need to provide sufficient context for review, respond to feedback and address problems identified in their submissions. Maintainers evaluate proposed changes, request revisions and decide whether integration is appropriate (Gousios et al., 2014). Contribution guidelines articulate expectations, while continuous integration checks can provide evidence for assessment (Elazhary et al., 2019). Together, these studies provide a rationale for expecting contributors to understand and explain their changes and to test and verify their behaviour: reviewers require assessable evidence, and accepted changes become part of a codebase requiring continued maintenance.

AI assistance makes this allocation of responsibility particularly relevant. Yang et al. (2026) identify policies requiring contributors to understand, verify and take responsibility for AI-assisted work. Such expectations extend established review responsibilities to how changes are produced. This provides a foundation for examining documented contributor obligations, human verification requirements and reviewer expectations without assuming that particular submissions involved AI or that stated responsibilities were fulfilled.

2.3 AI-assisted software development

Generative AI tools, such as GitHub Copilot, ChatGPT and Claude Code, increasingly support activities throughout the software development lifecycle, including code generation, debugging, testing and documentation. Dakhel et al. (2024) evaluated GitHub Copilot's code-generation capabilities, demonstrating both its potential and limitations in producing correct solutions. Similarly, Staron et al. (2024) reviewed applications of AI in testing and debugging, while Bhatia et al. (2024) investigated AI-generated unit tests, highlighting opportunities for automating tasks beyond code implementation.

However, empirical evidence regarding the productivity benefits of AI-assisted development remains mixed. Cui et al. (2024) reported increased task completion among developers using AI coding assistants in corporate field experiments. In contrast, Becker et al. (2025) found that AI assistance increased task completion time for experienced OSS developers working in familiar repositories. These contrasting findings suggest that productivity effects depend on developer experience, task characteristics and development context, rather than representing a consistent benefit across settings.

Collectively, this research demonstrates the breadth of AI assistance while cautioning against assuming that its adoption necessarily improves development outcomes. In collaborative OSS environments, the value of a contribution also depends on whether it can be understood, validated and maintained within established workflows. This shifts attention from individual productivity to the responsibilities surrounding AI-assisted work and motivates examining how projects communicate expectations and govern its use.

2.4 Quality, accountability, and review challenges

Assessments of AI-generated code indicate that its reliability cannot be inferred from plausible output alone. Nguyen and Nadi (2022) found variation in the correctness of Copilot suggestions

across programming languages and problems, while Cotroneo et al. (2025) identified different defect and vulnerability profiles in AI-generated and human-written code. These findings warrant scrutiny without implying that AI-produced contributions are uniformly inferior.

Evidence from repository-level use also highlights the importance of development context. Studying Cursor adoption in OSS projects, He et al. (2025) reported substantial but short-lived gains in development velocity alongside persistent increases in static analysis warnings and code complexity. Although these indicators do not directly establish subsequent maintenance failures, they suggest that faster code production can create additional quality-assurance demands. Long-term costs matter because software decisions can accumulate technical debt that requires later remediation (Kruchten et al., 2012).

In collaborative OSS workflows, these demands extend to reviewers and maintainers. Guizani et al. (2021) document challenges involved in contributing to large OSS organisations, illustrating the role of coordination and project practices in integrating changes. Contributors may therefore be expected to understand, test and explain AI-assisted work rather than treating generated output as self-verifying; maintainers must still assess whether changes are appropriate and supportable. Such responsibilities link quality concerns to human accountability (Yang et al., 2026).

Technical correctness is not the only concern. Surveying 574 developers, Stalnaker et al. (2026) documented varied views on copyright and ownership of AI-generated code, underscoring uncertainties relevant to licensing, attribution and provenance. A disclosure can make AI assistance visible to reviewers, but it neither verifies a contribution's security and maintainability nor establishes its licensing status. These distinct concerns help explain why OSS policies may combine transparency requirements with verification expectations, contributor responsibility statements and restrictions on particular uses of AI (Chen et al., 2026; Yang et al., 2026).

2.5 AI-use disclosure and its measurement in OSS

AI assistance affects both contribution and coordination in OSS. Song et al. (2025), using proprietary Copilot usage data alongside public repository records, found increased contributions but longer coordination times and more code discussion. This makes disclosure relevant to collaborative review, where maintainers must evaluate submitted changes. Within the pull-based review workflow examined by Gousios et al. (2014), PR descriptions offer a contribution-specific location for explanations and disclosure prompts.

Xiao et al. (2025) operationalise self-admitted GenAI use through explicit references in commit messages, code comments and documentation, combining automated retrieval with manual interpretation. Such evidence supports empirical analysis of reported tasks, purposes and governance expectations, linking declarations to inspectable artefacts. However, it captures visible reporting rather than all AI-assisted development: unreported assistance remains unobserved, and tool mentions require contextual interpretation. Consequently, self-reports depend on contributors' willingness to disclose and cannot independently verify whether their accounts are truthful.

Chen et al. (2026) measure AI self-disclosure through keyword matching in PR descriptions, including phrases such as "AI-assisted", and report counts and proportions of matching PRs. This provides a scalable indicator of reporting, but their described measure does not distinguish positive, negative, ambiguous and absent disclosures. In the present study, positive disclosures report assistance; negative disclosures explicitly deny AI use; ambiguous disclosures acknowledge AI-related requirements without resolving use or non-use; and no disclosure means no identifiable statement. Collapsing these categories would conflate reported assistance,

explicit non-use, uncertain responses and silence.

Peer-reviewed work on self-admitted technical debt demonstrates that text classification can mine developer declarations at scale, while keyword patterns may miss relevant statements (Maldonado et al., 2017). Applied to AI disclosure, wording, negation and template variation can produce false matches or omissions. Manual validation against human-coded records should therefore assess precision, the proportion of detected disclosures that are correct, and recall, the proportion of actual disclosure statements recovered. These metrics evaluate recognition of recorded statements, not detection of undisclosed AI use.

For RQ2, observable disclosure consequently supports analysis of how contributors respond to governance requirements, rather than a definitive compliance rate. An absent statement establishes neither undisclosed AI use nor a policy violation: AI may not have been used, the applicable rule may require disclosure only when it was used, or reporting may occur outside the PR description. Conversely, an identifiable statement may omit information required by the policy. PR descriptions therefore provide a bounded evidence source for comparing disclosure practices, interpreted alongside the scope and requirements of the policy in force.

2.6 AI governance policies, visibility, and enforcement

AI contribution policies govern the conditions under which assistance is acceptable, who remains responsible, and what evidence reviewers require. Chen et al. (2026) organise these provisions through TRACE: Transparency, Responsibility, Attribution, Constraints and Enforcement. Yang et al. (2026) instead distinguish prohibitionist, boundary-and-accountability, and quality-first orientations. These perspectives are complementary: TRACE separates policy requirements, whereas Yang et al.'s orientations explain how projects frame governance problems. Neither reduces governance to a binary decision about permitting AI.

Policy motivations also differ. Yang et al. (2026) identify review capacity, verification, accountability and provenance as connected concerns. Disclosure can make assistance visible for review; verification requirements shift testing and validation work to contributors to protect code quality; and accountability provisions require humans to understand and explain submitted changes. Rights and provenance concerns address whether material has an acceptable origin and can be redistributed, which technical testing cannot establish. Consequently, projects may prohibit uncertain inputs, permit assistance subject to disclosure and validation, or apply existing quality expectations regardless of tools. These approaches reflect different priorities rather than a universal progression towards stricter governance.

Chen et al. (2026) extend document analysis with matched longitudinal comparisons, reporting increased disclosure and review engagement following policy adoption. This offers evidence beyond policy statements, but their short observation window and reliance on quasi-experimental assumptions limit generalisation. Policy adoption also bundles several provisions, so these findings do not isolate policy placement or demonstrate consistent enforcement. Yang et al.'s qualitative account explicitly leaves causal effectiveness and uniform application unresolved.

Visibility is therefore a distinct consideration. Requirements may appear in external or organisation-level policies, repository documentation, contribution guidelines, or PR templates. Bringing a prompt into PR creation plausibly reduces the effort needed to discover it, but visibility does not establish that contributors read or understand it. Zhang et al. (2022) found that PR templates can structure information and support review, while contributors preferred essential description, testing and checklist content. However, this evidence concerns general contribution quality, not AI-disclosure compliance. Elazhary et al. (2019) also found substantial divergence between documented contribution processes and repository activity, cautioning

against equating written instructions with practice.

Policy visibility should also be distinguished from the strength of disclosure requirements. Chen et al. (2026) characterise transparency through the extent of required AI-use disclosure, ranging from voluntary declarations to detailed traceability. However, the detail required by a policy does not necessarily reflect how accessible it is to contributors. A comprehensive disclosure requirement may appear only in external documentation, while a PR template may prominently request minimal information. This distinction highlights the importance of considering both policy content and its integration into contribution workflows.

Governance also depends on how requirements are operationalised and enforced. PR templates can prompt declarations, reviewer guidelines can establish verification expectations, and automated checks can assess prescribed information. However, Elazhary et al. (2019) demonstrate that documented contribution processes do not necessarily reflect actual practices. Similarly, Chen et al. (2026) identify enforcement provisions ranging from requests for remediation to contribution rejection or participation restrictions. Such provisions indicate how projects intend to respond to non-compliance, but their existence does not establish whether or how consistently they are applied.

2.7 Research gap and study contribution

Existing research has examined how AI-assisted development influences OSS collaboration and how contributors report their use of generative AI (Song et al., 2025; Xiao et al., 2025). More recent studies have investigated the governance approaches adopted by OSS communities. Yang et al. (2026) identified differing governance strategies and concerns surrounding accountability, verification, and reviewer workload, while Chen et al. (2026) developed the TRACE framework and examined how AI policy adoption affects disclosure and developer experience. These studies establish an important foundation for understanding AI governance, but opportunities remain to examine how specific policy requirements are communicated, integrated into contribution workflows, and reflected in observable disclosure practices.

In particular, policy requirements, their visibility to contributors, and their implementation through contribution workflows represent related but distinct aspects of governance. Furthermore, aggregate disclosure measures may obscure differences between affirmative AI-use declarations, explicit statements of non-use, ambiguous responses, and absent disclosures. Since the absence of disclosure does not necessarily indicate AI use or a policy violation, observable reporting must be distinguished from verified compliance. Similarly, documented enforcement provisions do not establish whether maintainers consistently apply them.

This study complements existing research by combining manual governance coding with automated analysis of PR descriptions. It characterises contemporary policy approaches, contributor responsibilities, and underlying motivations (RQ1); examines observable disclosure patterns and their relationship with policy visibility (RQ2); and analyses documented workflow mechanisms and stated consequences (RQ3). By integrating these perspectives, the study provides a reproducible approach to comparing governance practices and recorded contributor behaviour. Its findings support descriptive analysis and exploratory associations rather than causal conclusions or direct measurement of undisclosed AI use and enforcement actions.

3. METHODOLOGY

3.1 Research design

This empirical study employed a mixed-methods approach, combining automated analysis of historical PR data with qualitative coding of OSS AI-use disclosure policies and governance practices. The methodology involved repository selection, automated disclosure detector development and validation, PR analysis, qualitative governance coding, and statistical analysis of policy visibility and disclosure rates.

3.2 Repository identification

Candidate repositories were identified using `melissawm/open-source-ai-contribution-policies`, an existing catalogue of OSS policies concerning AI-assisted contributions. As the catalogue's selection and curation methodology was not documented, it was used as an initial source of candidate repositories rather than an exhaustive or systematic sample. Entries classified as requiring AI-use disclosure were independently verified by examining the linked policy, contribution guidelines, and README, in that order. Only repositories with a current, identifiable disclosure requirement were retained.

3.3 Repository eligibility criteria

Following verification of the AI-disclosure requirement, candidate repositories were assessed against the predefined eligibility criteria presented in Table 1. Repositories were included only if they satisfied all criteria.

Table 1: Repository Selection Requirements

Criterion	Operational Requirement
Public Repository	Publicly accessible on GitHub
Open-source License	A recognised OSS license, such as MIT, Apache or GPL
Active Maintenance	More than 20 pull requests during the preceding 12 months
Pull-request workflow	Pull requests were used for accepting contributions
Non-trivial popularity	More than 500 GitHub stars
English documentation	Contributor-facing documentation was available in English
Disclosure requirement	The repository had a verifiable requirement to disclose AI use

38 repositories met all criteria, forming a purposive sample of projects with identifiable AI-disclosure requirements rather than a representative sample of OSS repositories.

3.4 Automated analysis tool

A purpose-built Java 17 and Maven command-line application was developed to collect and analyse PR data using the GitHub REST API. The tool automated the collection of historical PR data, identification of eligible contributions, detection of observable AI-use disclosures, and calculation of repository-level disclosure rates.

The analysis followed a sequential process. First, the tool retrieved closed PRs from each selected repository, removed duplicates, and applied predefined eligibility criteria. Next, it examined PR descriptions and visible text from their GitHub pages using a rule-based detector to identify observable AI-use disclosures. The tool then exported PR-level classifications and aggregated repository-level disclosure rates for subsequent statistical analysis. Automated classifications were subsequently evaluated against manually coded PRs to assess detector performance.

PR eligibility was determined using a rolling four-calendar-month window, calculated from the UTC timestamp at which each analysis began. Eligible PRs were closed, human-authored, and had a valid `closed_at` timestamp within this window. Both merged and unmerged PRs were included. The cutoff was calculated dynamically as `analysis_start_time_utc.minusMonths(4)`, with PRs closed before the cutoff or after the analysis timestamp excluded. Bot-authored contributions were excluded using GitHub account metadata and recognised bot naming patterns.

The application retrieved closed PRs through paginated, read-only GitHub REST API requests, using an access token supplied through an environment variable to increase the API request limit. Eligible PRs were sorted by `closed_at` descending, with PR number as a tie-breaker. No maximum PR count was imposed per repository, allowing all eligible PRs within the analysis window to be included.

For the final dataset, analysis began on 22 September 2026 at 09:58:43 UTC, establishing a cutoff of 22 May 2026 at the corresponding UTC time. The resulting dataset contained one record per eligible PR, including repository metadata, disclosure classification, and extracted disclosure evidence.

3.5 Automated disclosure classification

The tool employed a rule-based approach to automatically classify observable AI-use disclosures in PRs. The detector applied predefined text-matching patterns and structural rules to identify explicit statements of AI use or non-use. These rules were designed to accommodate variations in how contributors disclose AI assistance across OSS repositories.

The detector primarily analysed PR descriptions obtained through the GitHub REST API, supplemented by rendered webpage text as a fallback. Before classification, text was normalised by removing HTML comments, processing Markdown formatting, and excluding incomplete template prompts and other non-disclosure content. The detector then evaluated the text against rules recognising affirmative AI-use statements, explicit no-AI declarations, completed disclosure fields, checkbox responses, and contextual references to AI tools.

The detector incorporated both general-purpose and repository-specific rules. An inspection of the implementation identified 48 conceptual rule families, comprising 9 general semantic, 9 general structural, 3 tool/model vocabulary, 7 reusable template, 12 repository-specific, and 8 safeguard families. These counts represent distinct detection behaviours rather than individual regular expressions. General rules identified common disclosure expressions and structures, while repository-specific rules accounted for variations in PR templates and disclosure conventions across 12 repositories. Table 2 provides representative examples.

Table 2:Representative AI-use disclosure detection rules

Rule category	Example	Detection behaviour
General semantic	"I used ChatGPT"	Positive disclosure
General semantic	"No AI was used"	Negative disclosure
General structural	AI-disclosure heading followed by an answer	Interprets the response
Reusable template	`AIL: 0` or `AIL: 2`	Negative or positive disclosure
Repository-specific (QGIS)	AI-tool checkbox	Interprets checked and unchecked states
Repository-specific (Kornia)	Traffic-light AI-use options	Interprets template-specific selections
Safeguard	`CLAUDE.md` filename	Excludes non-disclosure references

A PR was classified as having a disclosure present when a qualifying positive, negative, or ambiguous AI-related declaration was identified. Positive disclosures explicitly acknowledged AI assistance, while negative disclosures explicitly stated that AI was not used. Ambiguous disclosures indicated that an AI-related declaration was present but did not clearly establish whether AI assistance had been used. All three categories were counted as detected disclosures when calculating the overall disclosure rate. PRs without a qualifying declaration were classified as disclosure absent.

This classification distinguished the observable presence of a disclosure from its meaning and completeness. In particular, an ambiguous disclosure did not necessarily satisfy a repository's disclosure requirements, while an absent disclosure did not definitively establish non-compliance, as valid declarations could remain undetected.

For each PR, the application recorded the disclosure classification, matched evidence, and evidence source. Comments, reviews, commit messages, linked issues, and check outputs were not systematically analysed.

3.6 Inter-rater reliability sample selection

An initial sample of 150 PRs was selected to assess inter-rater reliability between the two human coders responsible for identifying observable AI-use disclosures. This process aimed to establish consistency in manual classification before using human-coded labels to evaluate the automated detector.

To select the sample, duplicate entries were removed from the eligible repository list. The remaining repositories were shuffled using Java's `Collections.shuffle` method with a seeded `Random` instance. The first three repositories in the resulting order were selected without replacement: Homebrew/brew (R15), sympy/sympy (R38), and kubernetes/kubernetes (R20). Although the selection used a seeded pseudorandom process, the seed was not recorded, meaning the exact initial selection cannot be reproduced.

From each selected repository, the tool retrieved the 50 most recently closed eligible PRs. Eligibility criteria, including the rolling four-calendar-month window and exclusion of bot-authored contributions, were applied before the 50-PR limit. This produced an IRR sample of 150 PRs, with equal representation from the three repositories.

Each PR was assigned a stable identifier combining its repository name and PR number, allowing independent coder labels, consensus labels, and automated classifications to be matched reliably across datasets.

3.7 Inter-rater reliability

We independently coded the same 150 pull requests using an agreed framework to create human reference labels. A PR was coded as disclosure present when its description contained an observable response to the AI-use requirement, including an affirmative AI-use statement, an explicit no-AI declaration, or a completed disclosure field. Otherwise, it was coded as disclosure absent, indicating only that no observable disclosure was found.

Labels were matched using each PR's repository-and-number identifier. We agreed on all 150 classifications, so no disagreement resolution was required, and both sets of labels formed the consensus reference.

Inter-rater reliability was evaluated using Cohen's kappa:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \tag{1}$$

where P_o is the observed proportion of agreement and P_e is the agreement expected from our marginal label distributions.

We achieved 100% observed agreement and Cohen's $\kappa = 1.00$, indicating perfect agreement in applying the coding framework to the initial sample. This result demonstrates consistency between us human coders; it does not measure the accuracy of the automated detector.

3.8 Expanded detector evaluation

Initial detector evaluation revealed limitations in recognising AI-use disclosures across repositories, particularly where projects employed distinct disclosure formats, such as structured `AIL:` fields. To address these limitations, the evaluation strategy was expanded to include manually reviewed PRs from across the selected repositories. This enabled the identification of repository-specific disclosure conventions and informed subsequent refinement of the detector.

Up to 20 eligible PRs were randomly selected from each repository using a reproducible sampling procedure. The sampler retrieved closed PRs within a predefined four-calendar-month UTC window and excluded PRs previously selected for IRR analysis. Eligible candidates were sorted by PR number before being shuffled using Java's `Random` with a fixed seed of `42`. PRs were then selected from the shuffled order until 20 eligible PRs were obtained per repository, or no further eligible PRs remained. This produced an expanded sample of 708 PRs, separate from the initial 150-PR IRR sample.

The expanded sample aimed to meet or exceed the minimum of 381 observations required for a 95% confidence level and ±5 percentage-point margin of error under standard random-sampling assumptions, while also ensuring representation across repositories through a stratified sampling approach. Although this minimum was exceeded, the sampling strategy prioritised coverage across repositories rather than proportional representation of the overall PR population.

Each sampled PR was manually classified for the presence or absence of an observable AI-use disclosure. These labels were used to identify detection errors and refine general-purpose and repository-specific rules. Following refinement, automated classifications were compared against the manually reviewed labels using repository and PR number as stable identifiers. Results were categorised as true positives, true negatives, false positives, or false negatives to calculate detector performance metrics.

Following rule refinement, the detector's classifications were evaluated against the manually reviewed labels. Performance was assessed using accuracy, precision, recall, specificity, and F1-score, based on the resulting counts of true positives, true negatives, false positives, and false negatives. As the reviewed observations were also used to inform rule refinement, this evaluation represents performance on the development sample rather than independent validation on unseen data. The resulting performance metrics are reported in Section 4.1.

3.9 Full repository analysis

Following detector development and evaluation, the application analysed the 38 repositories listed in Appendix A, collecting all eligible PRs within the rolling four-calendar-month window. Unlike the initial sampling procedures, no maximum number of PRs was imposed per repository, allowing repositories to contribute observations according to their activity levels.

The final dataset contained 41,338 eligible PRs across all 38 repositories, collected between 22 and 23 September 2026. For each PR, the application recorded repository information, PR metadata, disclosure classification, and supporting evidence. These classifications were subsequently aggregated to calculate repository-level disclosure rates and examine patterns of observable AI-use disclosure across the selected repositories.

3.10 Visibility score

Each repository's AI-use disclosure policy was manually assigned a visibility score representing how likely a contributor was to encounter the policy during the normal contribution workflow. The score measured policy discoverability rather than the policy's strictness, comprehensiveness or enforcement.

Table 3: Visibility Scoring

Score	Visibility	Operational Definition
0	No policy found	No applicable AI-use disclosure policy could be located.
1	Low visibility	The policy appeared only in an external documentation site, separate repository or organisation-wide policy page that contributors would need to actively locate.
2	Moderate visibility	The policy was documented within the repository, such as in `CONTRIBUTING.md`, `AI_POLICY.md` or a repository documentation directory, but was not presented during PR creation.
3	High visibility	The policy was integrated directly into the contribution workflow, such as through a PR template.

Repositories were assigned the highest applicable visibility score based on the policy's most direct location. For example, a policy included in both repository documentation and a PR template received a score of 3. Scores were assigned through manual inspection using the operational definitions above; ambiguous or unavailable information was recorded as missing rather than scored. Although the rubric included score 0 for repositories with no identified policy, none of the selected repositories received this score.

3.11 Manual governance coding

Manual content analysis was conducted to examine how OSS projects governed AI-assisted contributions. The two project researchers reviewed the available AI-use policies, contribution guidelines, pull request templates, and relevant linked materials together. They discussed each repository and agreed on the coding. Because coding was collaborative rather than independent, no separate independent check or inter-rater reliability statistic was calculated.

The coding framework was developed for this study, informed by TRACE and the research questions. The researchers selected the governance themes they considered most relevant and created study-specific coding fields. A coding rubric provided a brief operational definition and inclusion or exclusion guidance for each field. It distinguished, for example, the detail a policy required contributors to disclose—from a basic AI-use statement to naming the tool, explaining its use, or providing traceability—and the location where disclosure was required. The framework also recorded workflow prompts, contributor responsibilities, attribution expectations, restrictions on AI use, verification mechanisms, and stated consequences. A prompt was not treated as evidence of verification, and consequences were coded only when expressly stated.

Coding was based on evidence in the reviewed materials. Wording such as "must" or "required" was treated as mandatory; encouragement to disclose was not coded as mandatory unless the surrounding wording made the obligation clear. Implicit rationales or practices were not inferred. When a field was not addressed in the materials, it was coded as "not stated" or "none found in examined materials," rather than taken as proof that the project had no such practice. Uncertain cases were discussed jointly and resolved using the rubric.

Multiple codes were allowed where the framework treated categories as independent or a policy stated several things. Each expressed rationale could be coded, and disclosure requirements for pull requests and commit messages were recorded separately because both could apply. Disclosure content was also recorded using separate fields, allowing a policy to require, for example, both the tool name and an explanation of its use.

Policies were first assessed in May 2026 during repository selection and reassessed in September 2026 for the governance analysis. Historical versions were not systematically reviewed; coding reflects the materials available during those assessments.

3.12 Statistical analysis

For each successfully analysed repository, the observed disclosure rate was calculated as:

$$Observed\ disclosure\ rate_i = \frac{D_i}{N_i} \tag{2}$$

where D_i is the number of detected disclosures and N_i is the number of successfully analysed eligible PRs. Failed analyses were excluded rather than counted as non-disclosures. Unretrieved

repositories were also excluded rather than assigned a rate of zero. These rates represent detector-observed disclosure, not definitive AI use or compliance.

Repository identifiers were normalised to account for differences in URLs, case, and trailing slashes before the analysis and policy datasets were matched.

Spearman's rank-order correlation was used because visibility was measured on an ordinal 0–3 scale. The hypotheses were:

$$H_o: \rho_s = 0 \quad (3)$$
$$H_1: \rho_s \neq 0 \quad (4)$$

where p_s represents the population Spearman correlation between policy visibility and observed disclosure rate. A two-sided significance test was used with a threshold of $\alpha = .05$. Each repository was treated as one equally weighted observation, irrespective of the number of eligible PRs it contributed. Governance scores were excluded because their criteria were not sufficiently developed for defensible statistical analysis.

3.13 Data analysis

The collected data were analysed using quantitative and qualitative descriptive techniques. At the pull-request level, PRs were classified as containing a positive disclosure (reported AI use), negative disclosure (explicitly reported non-use), or ambiguous disclosure (insufficient information to determine AI use). These classifications were aggregated by repository to examine the distribution of disclosure types and describe variation in observed disclosure rates.

The association between repository-level policy visibility scores and observed disclosure rates was examined using the procedure described in Section 3.12. This analysis was exploratory and assessed whether repositories with more visible disclosure requirements tended to have higher observed disclosure rates.

At the repository level, manually coded governance characteristics were summarised using frequency counts. These included disclosure requirements, contributor responsibilities, attribution expectations, AI-use constraints, enforcement mechanisms and stated consequences. Policy rationale categories were also counted to identify concerns frequently expressed in repository documentation. Because a repository could exhibit multiple characteristics or rationales, these categories were not treated as mutually exclusive.

Quantitative findings were interpreted alongside the manual governance coding to describe how documented practices related to observed disclosure behaviour. Selected policy excerpts illustrated differences in how repositories communicated and operationalised their requirements. These comparisons were descriptive and exploratory; the presence of a documented governance mechanism does not establish that it directly influences contributor behaviour.

4. QUANTITATIVE RESULTS

4.1 Detector performance on the development sample

The automated detector was evaluated against human reference labels for 708 manually reviewed PRs, randomly sampled across the selected repositories as described in Section 3.8. The resulting confusion matrix is presented in Table 4.

Table 4: Detector-Validation confusion matrix

	Human disclosure present	Human disclosure absent
Detector positive	278	1
Detector negative	4	425

The detector correctly identified 278 disclosures and 425 non-disclosures, producing one false positive and four false negatives. The corresponding performance metrics are presented in Table 5.

No ambiguous disclosures were identified in the 708-PR development sample. All identified disclosures were therefore classified as either positive or negative, both of which counted as disclosure present. Consequently, no separate treatment of ambiguous disclosures was required for this evaluation.

Table 5: Detector-performance metrics

Metric	Result
Accuracy	99.3%
Precision	99.6%
Recall	98.6%
Specificity	99.8%
F1 Score	99.1%

The performance metrics were calculated from the confusion matrix using true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

$$Specificity = \frac{TN}{TN + FP} \quad (8)$$

$$F_1 = 2\left(\frac{Precision \times Recall}{Precision + Recall}\right) \quad (9)$$

For this evaluation, $TP = 278$, $TN = 425$, $FP = 1$, and $FN = 4$.

The detector demonstrated high performance across all five metrics. Precision of 99.6% indicates that nearly all automatically detected disclosures were confirmed by human review, while recall of 98.6% indicates that the detector identified the vast majority of manually recognised disclosures. The low number of false positives and false negatives suggests that the

refined rule set effectively recognised the disclosure formats represented in the evaluated repositories.

However, these results should be interpreted in the context of the detector's iterative development. The manually reviewed sample was used to identify classification errors and refine detection rules, including repository-specific patterns. Consequently, the reported performance reflects the detector's fit to the reviewed sample and does not independently establish its accuracy on unseen repositories or disclosure formats.

4.2 Repository disclosure results

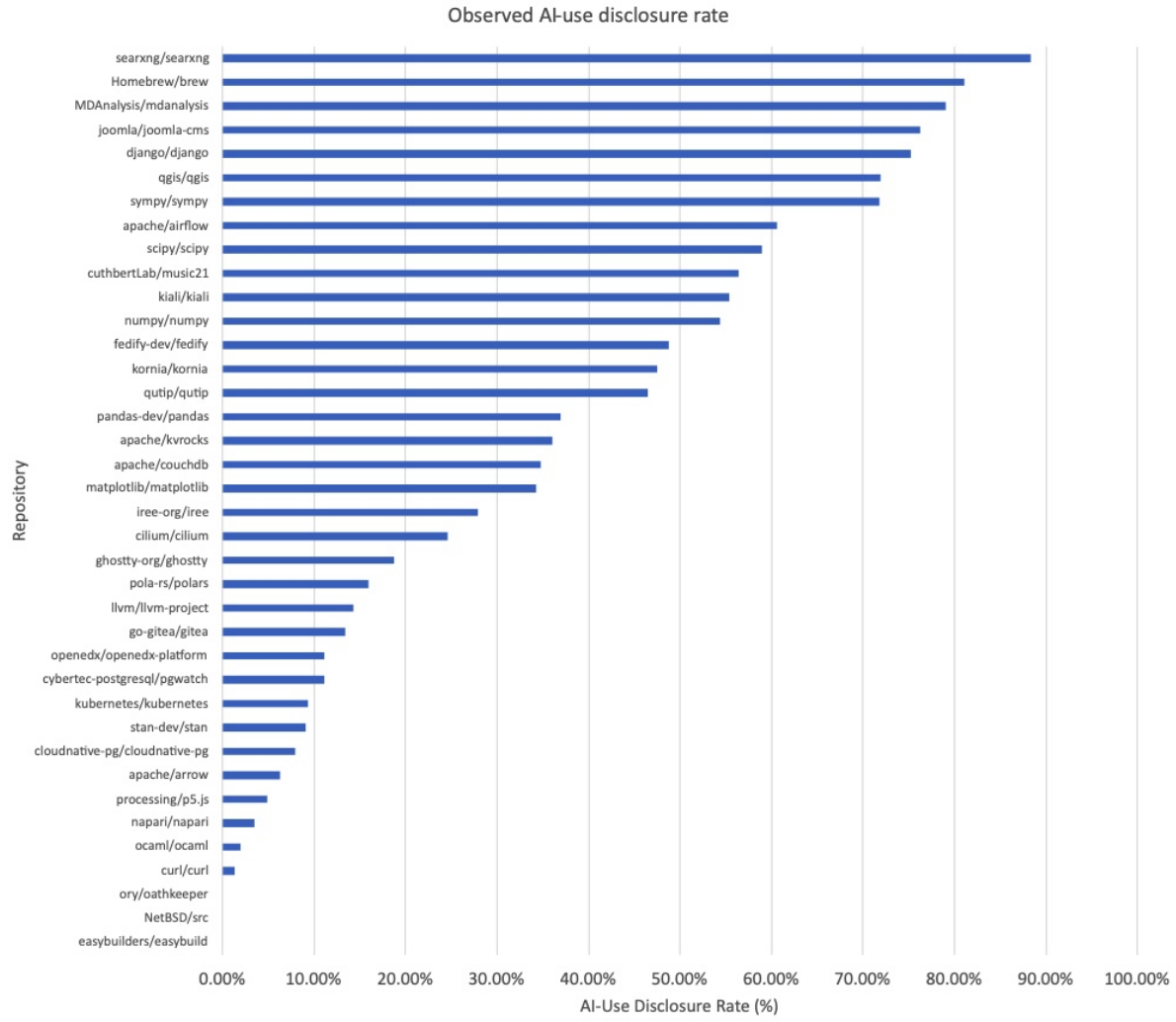
The final analysis included 38 open-source software repositories, comprising 41,338 eligible PRs collected within the four-month analysis window. Of these, 11,598 PRs (28.1%) contained an observable AI-use disclosure, while the remaining 29,740 PRs (71.9%) had no qualifying disclosure detected.

This pooled figure is dominated by one project. LLVM contributed 18,748 PRs (45.4% of the dataset) with a disclosure rate of 14.3%. Excluding LLVM, the pooled rate is 39.5%. Repository-level rates are therefore more informative than the overall figure (Figure 1).

Rates ranged from 0.0% to 88.3%. The highest were searxng/searxng (R36, 88.3%), Homebrew/brew (R15, 81.1%) and MDAnalysis/mdanalysis (R23, 79.1%). In total, 12 repositories exceeded 50% and 11 fell below 10%. Among larger projects, low rates included curl/curl (R07, 1.4% of 1,073 PRs), ocaml/ocaml (R27, 2.0% of 200) and napari/napari (R24, 3.5% of 312).

Three repositories recorded no detected disclosures: easybuilders/easybuild (R11), NetBSD/src (R25) and ory/oathkeeper (R29). However, they contributed only 1, 4 and 1 eligible PRs respectively, so these zeros say little about each project’s usual practice.

Figure 1: AI-Use Disclosure Rates Across OSS Repositories



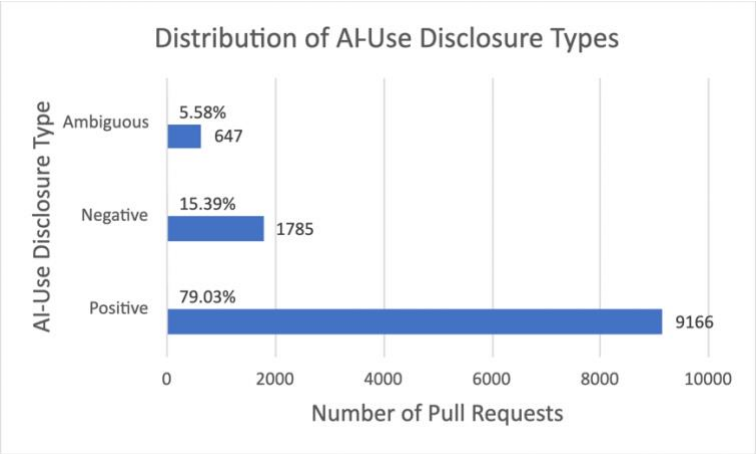
These findings demonstrate substantial variation in observable AI-use disclosure rates across the selected repositories, suggesting differences in contributor disclosure practices. However, the absence of a detected disclosure does not necessarily indicate non-compliance, as contributors may not have used AI in their contributions.

4.3 Types of AI-use disclosures

In addition to identifying whether an AI-use disclosure was present, the automated analysis distinguished between positive, negative, and ambiguous disclosures. Positive disclosures explicitly acknowledged the use of AI assistance, while negative disclosures explicitly stated that AI was not used. Ambiguous disclosures indicated the presence of an AI-related declaration without sufficient information to determine whether AI assistance had been used. All three categories contributed to the overall detected disclosure rate.

Of the 11,598 detected disclosures, 9,166 (79.0%) were classified as positive, 1,785 (15.4%) as negative, and 647 (5.6%) as ambiguous, as illustrated in Figure 2. Positive disclosures therefore accounted for the majority of detected declarations across the analysed repositories.

Figure 2: Distribution of AI-Use Disclosure Types Across OSS Pull Requests

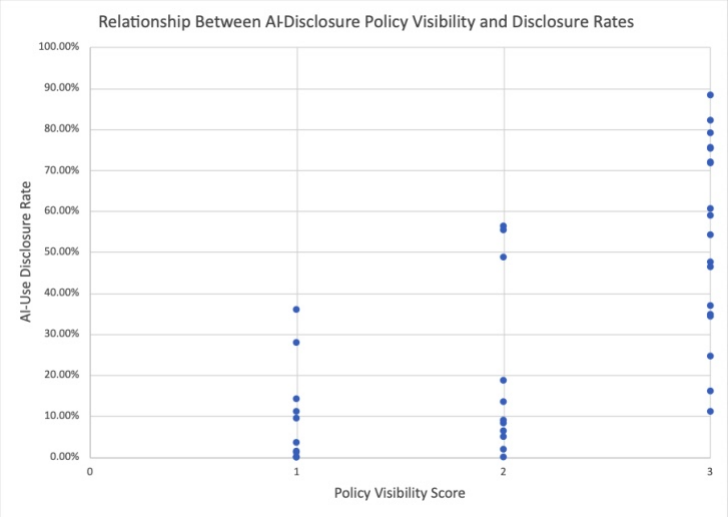


These findings highlight the distinction between the presence of an AI-use declaration and an affirmative acknowledgement of AI assistance. Although 28.1% of eligible PRs contained a detected disclosure, this included explicit statements of non-use and ambiguous declarations. Consequently, the overall disclosure rate should not be interpreted as the proportion of contributions developed with AI assistance.

4.4 Visibility and disclosure rate

The relationship between AI-use disclosure policy visibility and observed disclosure rates was examined across the 38 repositories. Policy visibility was assessed using the four-level rubric described in Section 3.10, ranging from 0 (no identifiable policy) to 3 (disclosure requirements integrated into pull request templates). Figure 3 illustrates the relationship between policy visibility scores and repository-level disclosure rates.

Figure 3: Relationship Between AI-Disclosure Policy Visibility and Disclosure Rates



Spearman's rank-order correlation identified a statistically significant positive association between policy visibility and observed disclosure rates ($\rho = .618$, $p < .001$, $n = 38$). This indicates a moderately strong positive monotonic relationship, with repositories that integrated disclosure requirements more directly into their contribution workflows generally exhibiting higher observed disclosure rates.

However, the relationship was not uniform across repositories, indicating that policy visibility alone does not fully explain differences in disclosure behaviour. Other factors, including repository culture, policy wording, contributor practices, and maintainer expectations, may also influence whether contributors provide explicit AI-use declarations.

Furthermore, the observed association does not establish a causal relationship between policy visibility and disclosure behaviour. The visibility rubric measures the accessibility and workflow integration of policy requirements rather than contributors' actual awareness of or adherence to those requirements. Therefore, these findings suggest that greater policy visibility is associated with higher observed disclosure rates, but do not establish that increasing policy visibility would independently improve disclosure compliance.

5. GOVERNANCE ANALYSIS RESULTS

5.1 Governance requirements and practices

Manual governance coding was conducted across 38 OSS repositories to examine how projects regulate AI-assisted contributions. The analysis identified variation in disclosure requirements, contributor responsibilities, attribution expectations, and restrictions on AI use.

Disclosure requirements differed in the level of detail expected from contributors. Of the 38 repositories, 13 required identification of the AI tools used, while 12 required explanations of how AI contributed to the work. A further three required more detailed traceability, whereas seven specified only basic disclosure. Requirements were also integrated into contribution workflows differently, with 21 repositories incorporating AI-related requirements into pull request templates.

Policies also differed in where disclosures were explicitly required. Twenty-two repositories specified that contributors must disclose AI use within their pull requests, while four explicitly required disclosure in commit messages. PR-level disclosure was recorded only where a policy explicitly identified the pull request as the required location for disclosure. General requirements to disclose AI use, without specifying where the declaration should appear, were not classified as PR-level requirements. This distinction separates the obligation to disclose AI assistance from the location in which that disclosure must be made.

Contributor responsibility was another prominent governance dimension. Most repositories (24/38) explicitly required contributors to review or test AI-assisted work, while eight required contributors to explain their contributions and four specified general accountability expectations. In contrast, attribution requirements were less consistently documented, with 21 repositories having no explicit attribution requirements and 14 specifying licensing or Developer Certificate of Origin (DCO) obligations.

Restrictions on AI use were also common. Nineteen repositories imposed restrictions in specific contexts, while three explicitly restricted bulk or low-effort AI-generated contributions. One repository prohibited AI-generated contributions more generally. These findings demonstrate that OSS governance extends beyond disclosure, with projects establishing varying expectations for contributor responsibility, provenance, and acceptable AI-assisted development practices.

A further governance requirement concerned the use of AI in contributor–maintainer communication. Of the 38 repositories, 29 (76.3%) explicitly restricted or prohibited AI-generated communication during interactions with reviewers or maintainers. For example,

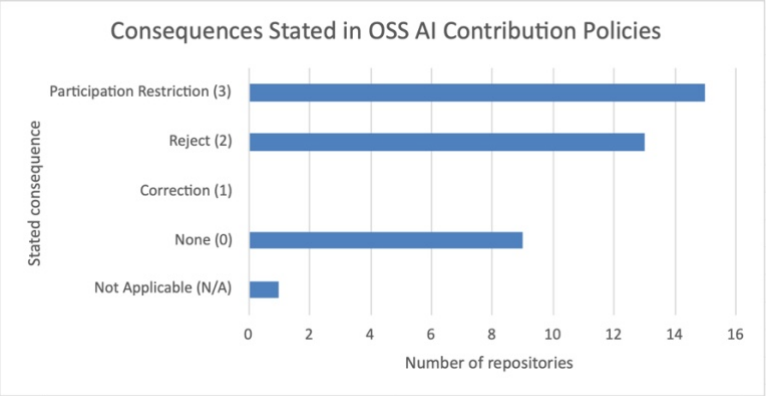
Cilium (R05) prohibited substantially AI-generated messages in community discussions, while SciPy (R35) and SymPy (R38) emphasised the importance of human-to-human communication, allowing exceptions for translation and grammar editing. These restrictions demonstrate that AI governance extends beyond code quality and disclosure to preserving direct human engagement, contributor accountability, and the integrity of collaborative review processes.

5.2 Enforcement mechanisms and stated consequences

The governance analysis also examined the consequences repositories explicitly specify for violations of their AI-use policies. These ranged from no documented consequences to pull request rejection and restrictions on future participation.

Of the 38 repositories, 15 specified participation restrictions as a potential consequence, while 13 explicitly identified pull request rejection or closure. Nine repositories did not state any consequences, and one described enforcement as being at the maintainers' discretion without specifying a particular action.

Figure 4: Distribution of Stated Consequences Across OSS Repositories



These findings indicate that many OSS projects communicate consequences for non-compliance, although the severity and specificity of enforcement vary. While some repositories establish explicit grounds for rejecting contributions, others describe broader consequences for repeated violations. However, these results represent documented enforcement provisions rather than observed maintainer actions and therefore do not establish how consistently such consequences are applied in practice.

5.3 Motivations behind AI governance policies

The manual governance coding identified several recurring motivations underlying AI-use policies, including reviewer workload, human accountability, intellectual property rights and provenance, code quality, and transparency. Multiple rationales could be recorded for each repository, reflecting the range of concerns associated with AI-assisted contributions. Repository identifiers correspond to those presented in Appendix A, Table A.1.

Reviewer workload was the most frequently identified rationale, appearing in 20 of the 38 repositories. This concern reflects the additional maintenance burden that AI-assisted contributions may introduce, particularly when contributors submit large volumes of generated code without sufficient understanding or verification. Such contributions can require maintainers to spend additional time reviewing, debugging, and requesting revisions, potentially diverting resources from other development activities. Relatedly, code quality was

identified in 13 repositories, reflecting concerns about the correctness, reliability, and maintainability of AI-generated code. Although these rationales overlap, reviewer workload focuses on the resources required to evaluate contributions, whereas code quality concerns the technical standard of the submitted work.

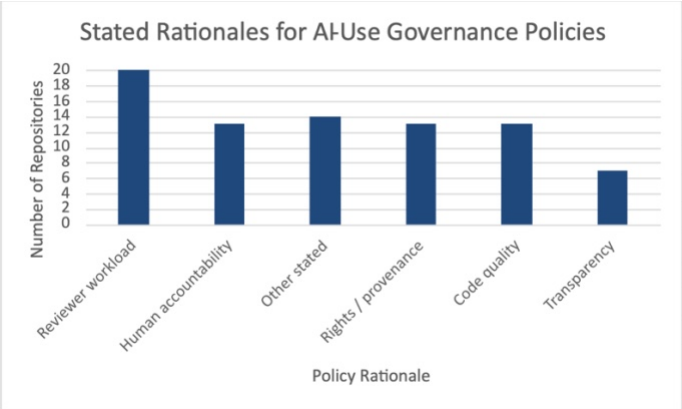
Human accountability was identified in 13 repositories, emphasising that contributors remain responsible for the code they submit, regardless of whether AI tools were involved. Policies expressing this rationale commonly expect contributors to understand, review, and test AI-generated content rather than relying on its apparent correctness. This reinforces the expectation that AI assistance does not diminish individual responsibility for contributions.

Rights and provenance concerns appeared in 13 repositories, reflecting uncertainty surrounding the ownership, licensing, and origins of AI-generated material. These policies seek to ensure that contributions comply with existing intellectual property and licensing requirements, particularly where AI-generated content may reproduce or derive from third-party material. Provenance concerns also relate to the ability to establish where contributed code originated and whether contributors have the necessary rights to submit it.

Transparency was explicitly identified as a rationale in seven repositories. This reflects the importance of making AI involvement visible to maintainers and reviewers, allowing them to understand how contributions were developed and assess whether additional scrutiny may be appropriate. Although disclosure requirements were common across the repositories examined, transparency was less frequently stated as an explicit motivation than concerns relating to maintenance burden, accountability, and provenance.

Beyond the predefined categories, 14 repositories identified additional motivations for AI-use governance. Several raised concerns about the reliability of AI-generated content, with Airflow (R01) highlighting security and stability risks associated with uncritically adopting generated code, while Django (R10) and Joomla (R17) noted the potential for inaccurate or fabricated information. Other repositories emphasised maintaining meaningful contributor engagement and healthy community practices. Arrow (R02) highlighted collaborative learning between contributors and reviewers, while Gitea (R14) expressed concerns about contributors relying excessively on AI rather than engaging substantively in the review process. Broader ethical considerations were also evident, including EasyBuild's (R11) concerns about the training data used by large language models and p5.js's (R32) consideration of environmental impacts. Stan (R37) additionally emphasised the importance of maintaining community trust. Overall, these findings suggest that OSS projects introduce AI-use policies not solely to encourage disclosure, but also to preserve technical reliability, support meaningful human participation, maintain community trust, and address broader ethical uncertainties surrounding AI-assisted development.

Figure 5: Stated Motivations for AI-Use Governance



6. DISCUSSION

Our findings show that the selected open-source software projects addressed AI-assisted contributions through requirements for disclosure, human verification, acceptable use, and consequences for policy violations. These requirements connect transparency with the practical work of reviewing and maintaining software. By examining both policy documents and PRs, this study provides evidence about how projects communicate their expectations and what contributors record when submitting changes. The findings also show why policy requirements, recorded disclosure, and actual compliance must be interpreted separately.

6.1 Interpreting disclosure and compliance

The analyser identified disclosure evidence in 11,598 of 41,338 PRs (28.1%), with repository-level rates ranging from 0% to 88.3%. This variation indicates that contributors recorded AI-related information very differently across the selected projects. A single overall rate therefore provides an incomplete account of disclosure practices.

The 28.1% figure measures the presence of evidence recognised by the detector. The detection rules included statements that AI was used, statements that it was not used, and some acknowledgements of policy compliance. Consequently, the result does not mean that 28.1% of contributions used AI. Similarly, a PR without a detected disclosure does not establish that its author used AI without reporting it. Some contributors may not have used AI, and some policies require disclosure only when particular uses occur.

Policy scope further limits a uniform interpretation of compliance. The governance sheet recorded a PR-level disclosure requirement for 22 of the 38 projects, while other policies concerned different contribution channels or restrictions on AI use. Assessing compliance requires establishing which rule applied to a particular contribution and whether the contributor satisfied it. Our results therefore address RQ2 through observable reporting practices, but cannot establish the proportion of AI-assisted contributions that violated mandatory disclosure rules.

This distinction is consistent with Xiao et al. (2025), who studied explicit admissions of GenAI use and acknowledged that their evidence captured only visible use. Our measure covers a broader range of AI-related statements, including non-use, so the two studies' counts are not directly comparable. Their work nevertheless reinforces the need to distinguish recorded statements from the underlying use of AI.

6.2 Policy visibility and contribution workflows

Policy visibility was positively associated with observed disclosure rates (Spearman's $\rho = .618$, $p < .001$, $n = 38$). Repositories with policies integrated into PR templates generally recorded higher disclosure rates than repositories whose policies appeared in less immediate locations. One plausible explanation is that a prompt at submission makes the requirement easier to encounter and gives contributors a clear opportunity to respond. Taken together with the 21 policies incorporating AI-related requirements into PR templates, this association suggests that policy placement forms part of how projects operationalise governance: the contribution interface connects written expectations with an opportunity to respond.

However, visible prompts did not produce uniformly high disclosure rates. Differences in contributor familiarity, maintainer practices, policy wording, and the kinds of contributions received may also explain the variation. Structured templates may additionally make disclosures easier for the analyser to recognise. The observed relationship could therefore reflect both differences in contributor behaviour and differences in measurement. Our comparison does not establish that moving a policy into a template causes disclosure to increase.

Chen et al. (2026), in Making AI Visible, Not Vanished, reported increased disclosure and review engagement following AI-policy adoption using a longitudinal comparison with matched repositories. Our findings complement this work by examining where contributors encounter policies. Their TRACE framework describes transparency partly through the information a policy requires, whereas our visibility score describes its location within the contribution workflow. These are distinct considerations: a detailed requirement may be difficult to find, while a prominent checkbox may request little useful information. Effective policy design needs to consider both accessibility and the substance of the disclosure.

6.3 Human responsibility and stated consequences

The governance results indicate that projects commonly made human verification an explicit condition of AI-assisted contribution. Twenty-four of the 38 policies required contributors to review or test their work, while eight required them to explain it. These expectations identify the contributor as the person responsible for understanding and supporting the submitted change. Disclosure gives reviewers context, but the requirement to verify and explain a contribution addresses whether someone can respond meaningfully when problems arise.

Projects also described different consequences for unacceptable contributions. Fifteen policies included restrictions on future participation, and thirteen identified rejection or closure as their highest coded consequence. These provisions give maintainers grounds to refuse work or respond to repeated problems. However, some consequences applied to poor quality, disruptive conduct, or failure to verify a contribution, rather than specifically to missing disclosure. Their presence therefore cannot be interpreted as evidence that projects routinely penalised nondisclosure.

Nine policies stated no explicit consequence, and one used wording that could not be assigned confidently to the consequence categories. This does not demonstrate an absence of enforcement: maintainers may act through review practices or broader community rules. Conversely, a written sanction does not demonstrate that maintainers applied it consistently. The study addresses the documented mechanisms in RQ3 by identifying requirements and available responses. Establishing how projects actually enforce them requires systematic analysis of review comments, closure decisions, and subsequent contributor behaviour.

6.4 Motivations underlying AI governance

Reviewer workload was the most frequently stated rationale, appearing in 20 of the 38 projects. This finding suggests that a central concern is the effort required to assess incoming contributions. Producing a plausible change does not establish that the change is necessary, correct, or suitable for the project. When contributors submit work they have not checked or cannot explain, reviewers may inherit that verification effort.

This interpretation aligns with Yang et al. (2026), whose analysis of governance materials from 67 OSS project and policy cases identified review capacity, accountability, verification, and provenance as recurring concerns. Our results add evidence about how frequently selected motivations were explicitly stated within our policy sample. They support an interpretation of AI governance as a response to the distribution of work between contributors and maintainers.

The other rationales clarify different aspects of that response. Code-quality concerns address whether a contribution works correctly and can be maintained. Human-accountability concerns address who understands the change and will explain or correct it. Rights and provenance concerns, recorded in 13 projects, address the origins of contributed material and whether its inclusion is compatible with the project's licensing and contribution requirements. These concerns can overlap, but each points to a different reason for requiring information or verification from contributors.

Only seven projects explicitly stated transparency as a rationale. This suggests that disclosure often serves wider governance objectives, such as supporting review or documenting responsibility. However, the counts represent stated motivations, not a ranking of everything maintainers value. A project may require disclosure without explaining why, and multiple motivations could be coded for the same project. The policy analysis also does not establish that the stated risks occurred in the analysed PRs or that the policies successfully reduced them.

6.5 Limitations of the study

The analyser enabled the study to examine disclosure evidence across a substantially larger collection of PRs than the manually reviewed sample. Its rules and recorded evidence also make individual decisions inspectable. The evaluation reported 99.3% accuracy across 708 matched records, supporting its usefulness for recognising the disclosure formats represented in that sample. Several limitations nevertheless constrain interpretation of the results.

This performance does not establish equivalent accuracy in unfamiliar repositories. The detector combined reusable language and formatting rules with explicit rules for 12 repositories, and its development included refinements following manual error review. The evaluation sample was used to refine the detector, so the reported performance represents its fit to the development sample rather than independent validation. Sampling up to 20 PRs per repository prioritised coverage of disclosure conventions over proportional representation of all PRs. The resulting metrics should not therefore be treated as a population-wide accuracy estimate with a guaranteed ±5 percentage-point margin of error. Testing on repositories and later PRs excluded from rule refinement would provide stronger evidence of generalisability. Comparing versions with and without repository-specific rules would also clarify how much performance depends on local configuration.

Some measurement limits remain even when the detector reproduces manual labels accurately. Acknowledging a policy does not necessarily identify whether AI was used, and an unanswered checkbox may be difficult to distinguish from an intended negative response. The analysis primarily examined PR descriptions, with rendered webpage text as a fallback; it did not

systematically capture disclosures made only in commits, reviews, or subsequent discussion. This matters because four projects explicitly required disclosure in commits. A stronger measurement approach would distinguish policy acknowledgements from unresolved responses within the ambiguous category, while checking the channels required by each policy. Even accurate classification of these statements cannot establish undisclosed AI use or whether a contribution satisfied every applicable rule.

The dataset also limits broader generalisation. The 38 repositories were purposively selected from an existing policy catalogue and do not represent all OSS projects. The catalogue's undocumented curation process and the eligibility criteria may favour particular projects. PR volumes also differed substantially. LLVM supplied approximately 45% of all analysed PRs, so the pooled disclosure percentage reflects activity in larger projects more heavily. At the other extreme, rates based on only a few PRs provide limited evidence about a project's usual practices. Repository-level comparisons should therefore accompany the overall percentage, and the number of PRs should not be confused with the number of independent policy cases.

The analysis included all eligible closed PRs within a rolling four-calendar-month window, without a per-repository cap. It therefore captures a limited period and excludes contributions still open at collection. Policy interpretation also depends on accurate repository matching, policy scope, and timing. A policy's current wording may differ from the requirements that applied when an earlier PR was submitted. Dated policy versions and checks of applicability would strengthen the connection between policy evidence and the behaviour being measured.

Written rationales and consequence categories require judgement, particularly where wording is implicit or ambiguous. We reviewed the governance policies jointly and agreed coding decisions through discussion. This supported a shared interpretation, but did not provide an independent assessment of the coding and cannot eliminate shared interpretive bias. The initial agreement on PR disclosure labels concerns a separate coding task. Moreover, documented consequences and rationales establish what projects state, rather than how consistently maintainers enforce policies or whether those policies reduce the stated risks.

Finally, the visibility–disclosure association does not establish causation. Repository culture, policy wording, contributor practices, and the detector's sensitivity to structured templates may influence the relationship. Each repository received equal weight in the correlation despite differing PR counts, so rates based on small samples could influence the result. The ordinal visibility score also measures policy location rather than contributors' actual awareness. These limitations support interpreting the association as exploratory evidence within the selected repositories.

6.6 Synthesis of answers to the research questions

Policy coding identifies what projects require and why, while the quantitative analysis shows what contributors record in PRs. Together, these components support the following answers.

RQ1: How are OSS projects adapting contribution governance practices in response to AI-assisted development?

The selected projects formalise expectations for disclosure, human verification, and acceptable AI use. Across the 38 policies, 24 required contributors to review or test their work and eight required them to explain it (Section 5.1). These requirements retain human responsibility for AI-assisted contributions and place disclosure within broader contribution standards.

RQ2: To what extent do contributors comply with mandatory AI-use disclosure policies in OSS pull requests?

The study establishes the extent of observable reporting: 11,598 of 41,338 PRs (28.1%) contained detected disclosure evidence, with repository rates from 0% to 88.3% (Section 4.2). Policy coding shows why this is only a partial answer to compliance: requirements differ in scope and reporting channel. Actual compliance cannot be quantified without knowing whether AI was used and which obligation applied.

RQ3: How do OSS projects operationalise and enforce AI-use disclosure policies within contribution workflows?

Projects operationalise requirements through submission prompts and documented consequences: 21 used PR templates, while 15 specified participation restrictions and 13 identified rejection or closure as their highest coded consequence (Sections 5.1–5.2). The positive visibility–disclosure association ($\rho = .618$) links these workflow arrangements to recorded reporting. It does not establish that prompts caused disclosure or that the stated sanctions were applied.

6.7 Practical implications and future research

The findings suggest several practical improvements for projects designing AI contribution policies. Maintainers can place a concise disclosure prompt where contributors submit work, explain which uses must be reported, and specify the information needed for review. Separate responses for AI use and non-use can reduce uncertainty about whether a field was intentionally completed. Where AI was used, a short explanation of its role and the contributor's verification may provide more useful context than a general acknowledgement alone.

Projects can also make responsibility and consequences easier to understand. Policies should explain what contributors must review, test, or be able to justify, and distinguish an incomplete disclosure from an unverified contribution or repeated disruptive behaviour. These recommendations follow from the variation documented in this study; their effects on workload, contribution quality, and participation still require evaluation.

For researchers, the study's contribution is the combined examination of policy visibility, documented governance mechanisms, and recorded disclosure within a defined repository sample. This connects what projects ask of contributors with what appears in submission records, while identifying the information needed to make stronger claims about compliance. Future research can extend this approach through observations before and after policy changes, systematic analysis of enforcement decisions, and interviews with contributors and maintainers. Evaluations should examine whether clearer reporting supports useful review and sustainable participation, alongside measuring how often contributors complete a disclosure.

7. CONCLUSION

This study combined automated analysis of 41,338 eligible pull requests across 38 open-source software repositories within a four-month window with manual coding of AI contribution policies. Observable disclosure was detected in 11,598 PRs (28.1%). This rate measures the proportion of analysed PRs containing detector-recognised disclosure evidence, including reported AI use, explicit statements of non-use, and ambiguous declarations. It therefore does not represent the prevalence of AI-assisted development or the proportion of contributions that complied with every applicable policy.

Policy visibility was positively associated with repository-level disclosure rates (Spearman's $\rho = .618$, $p < .001$). Repositories presenting disclosure requirements more directly within

contribution workflows generally exhibited higher observed disclosure rates. However, this association does not establish that greater visibility causes improved disclosure or compliance.

The governance analysis showed that AI-use policies extend beyond disclosure to contributor responsibility, acceptable use, and stated consequences. Twenty-four of the 38 repositories explicitly required contributors to review or test AI-assisted work, while reviewer workload was the most frequently stated rationale, appearing in 20 repositories. Together, these findings suggest that AI governance seeks to preserve contribution standards and manage review demands while maintaining human accountability.

The study provides a baseline for understanding how documented governance relates to recorded disclosure within the selected repositories. Its principal interpretive boundary is that public policies and disclosure statements cannot establish undeclared AI use, verified compliance, or actual enforcement. The purposive sample also limits generalisation to OSS more broadly. For maintainers, the findings support considering clear disclosure prompts alongside explicit verification expectations. Future research should evaluate how these practices affect disclosure, review effort, and participation over time, using independent detector validation and direct examination of enforcement decisions.

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Appendix A

Table A1: Repositories included in the study and their reference identifiers.

ID	Repository	Visibility Score	Eligible PRs Analysed	Disclosures Detected	Disclosure Rate
R01	apache/airflow	3	3952	2395	0.6060
R02	apache/arrow	2	598	38	0.0635
R03	apache/couchdb	3	89	31	0.3483
R04	apache/kvrocks	1	86	31	0.3605
R05	cilium/cilium	3	1746	431	0.2468
R06	cloudnative-pg/cloudnative-pg	1	264	21	0.0795
R07	curl/curl	1	1073	15	0.0140
R08	cuthbertLab/music21	2	108	61	0.5648
R09	cybertec-postgresql/pgwatch	3	81	9	0.1111
R10	django/django	3	594	447	0.7525
R11	easybuilders/easybuild	1	1	0	0.0000
R12	fedify-dev/fedify	2	121	59	0.4876
R13	ghostty-org/ghostty	2	632	119	0.1883
R14	go-gitea/gitea	2	1002	135	0.1347
R15	Homebrew/brew	3	1264	1025	0.8109
R16	iree-org/iree	1	283	79	0.2792
R17	joomla/joomla-cms	3	427	326	0.7635

ID	Repository	Visibility Score	Eligible PRs Analysed	Disclosures Detected	Disclosure Rate
R18	kiali/kiali	2	426	236	0.5540
R19	kornia/kornia	3	584	278	0.4760
R20	kubernetes/kubernetes	1	2062	194	0.0941
R21	llvm/llvm-project	1	18748	2681	0.1430
R22	matplotlib/matplotlib	3	448	154	0.3438
R23	MDAnalysis/mdanalysis	3	43	34	0.7907
R24	napari/napari	3	312	11	0.0353
R25	NetBSD/src	1	4	0	0.0000
R26	numpy/numpy	3	925	503	0.5438
R27	ocaml/ocaml	2	200	4	0.0200
R28	openedx/openedx-platform	1	287	32	0.1115
R29	ory/oathkeeper	2	1	0	0.0000
R30	pandas-dev/pandas	3	1537	569	0.3702
R31	pola-rs/polars	3	1067	171	0.1603
R32	processing/p5.js	2	183	9	0.0492
R33	qgis/qgis	3	677	487	0.7194
R34	qutip/qutip	3	71	33	0.4648
R35	scipy/scipy	3	749	442	0.5901
R36	searxng/searxng	3	326	288	0.8834

ID	Repository	Visibility Score	Eligible PRs Analysed	Disclosures Detected	Disclosure Rate
R37	stan-dev/stan	2	22	2	0.0909
R38	sympy/sympy	3	345	248	0.7188